

Minkuk Kim

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Artificial Intelligence, Kyung Hee University (KHU)

CONTACT INFORMATION

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EDUCATION

Kyung Hee University

Ph.D. in Department of Artificial Intelligence

Aug. 2024 - Current
Yongin, Republic of Korea

- Advisor: Prof. Seong-Tae Kim

Kyung Hee University

M.S. in Department of Artificial Intelligence

Aug. 2022 – Aug. 2024
Yongin, Republic of Korea

- Advisor: Prof. Seong-Tae Kim

Kyung Hee University

B.S. in Department of Computer Science and Engineering

Mar. 2016 - Aug. 2022
Yongin, Republic of Korea

- Advisor: Prof. Seong-Tae Kim

RESEARCH KEYWORDS

Core Theme

Memory-Augmented Video Understanding

Research Scope

Long-form and Streaming Video Understanding with Video-Language Models

Representative Tasks

Video Question Answering, Temporal Video Grounding, Video Captioning

RESEARCH EXPERIENCE

Long-horizon Video Understanding

June. 2025 - Current

- Question-aware Key-frame Selection for Long-form Video QA (*Submitted to ECCV 2026*)
- We propose a question-aware key-frame selection framework for long-form video understanding.
- We use model confidence to guide re-thinking reasoning planning and key-frame selection.
- Structured Temporal Memory for Streaming Video Understanding (*Submitted to NeurIPS 2026*)
- We structured streaming observations into fine-grained temporal memory for long-horizon video streaming.
- We design an adaptive reasoning pipeline that includes strategy planning and evidence grounding.

Dense Video Captioning

June. 2023 - Current

- Dense Video Captioning with Cross-modal Memory Retrieval. (**CVPR 2024**)
- We propose a new dense video captioning model with cross-modal memory retrieval.
- We propose a novel structure that can learn cross-modal correlation.
- Hierarchical Memory Modeling for Dense Video Captioning. (**AAAI 2025**)
- We propose a new dense video captioning model with hierarchical memory retrieval.
- We propose a novel hierarchical compact memory structure that can save compact information in hierarchy.
- Hour-long Dense Video Captioning for Movie. (*Preparation for CVPR 2027*)
- We are working toward submission to the CVPR 2027 conference.
- We explore video localization and description for long-form video that can be considered with context on movie domain.

PUBLICATIONS

International Conference Papers

[1] **M. K. Kim**, H. B. Kim, J. Y. Moon, J. W. Choi[†] and S. T. Kim[†]

“Do You Remember? Dense Video Captioning with Cross-Modal Memory Retrieval.”
Computer Vision and Pattern Recognition (CVPR), Seattle, U.S.A., Jun. 2024.

[2] **M. K. Kim**, H. B. Kim, J. Y. Moon, J. W. Choi[†] and S. T. Kim[†]

“HiCM²: Hierarchical Compact Memory Modeling for Dense Video Captioning”
Association for the Advancement of Artificial Intelligence (AAAI), Philadelphia, U.S.A., Feb. 2025.

[3] S.H.Kim, **M. K. Kim**, H. B. Kim, J. W. Choi[†] and S. T. Kim[†]

“Retrieval-Augmented Natural Language Reasoning for Explainable Visual Question Answering”
International Conference on Image Processing Workshop (ICIPW), Abu Dhabi, U.A.E., Oct. 2024.

International Journal Papers

- [1] E. K. Cho, **M. K. Kim**, H. I. Kim, J. Y. Moon and S. T. Kim[†]
“Exploiting recollection effects for memory-based video object segmentation.”
Image and Vision Computing, 140, 2023.

PATENTS

- [1] **M. K. Kim**, E. K. Cho, J. U. Kim and S. T. Kim
“Deep neural network uncertainty prediction device and method.”
Korean Patent Application (10-2023-0110754), Aug. 23, 2023.

RESEARCH PROJECTS

- [1] Video Spatio-temporal Localization for Long-term memory analysis
· Participating Researcher (2023-2025)
· Supervision : **ETRI** (Electronics and Telecommunications Research Institute)
- [2] Research on Causality-centric Situation Understanding based on Short-Long Term Episodic Memory
· Participating Researcher (2026-Current)
· Supervision : **ETRI** (Electronics and Telecommunications Research Institute)

EXPERIENCES

- GPU Server System Administrator** (2023–2024)
· Management of Cluster servers that has 370 GPUs with DeepOps and Slurm.
- Teaching Assistant** (2023–Current)
· Web and Python Programming (SWCON104)
· Image Processing and Pattern Recognition (SWCON494-00)
· Object-oriented Programming (CSE103-02)

ACADEMIC SERVICES

- Conference Reviewer**
· AAAI Conference on Artificial Intelligence (AAAI)
· Neural Information Processing Systems (NeurIPS)

SKILLS

Language Skills

- Korean(native), English

Programming Skills

- Python, C/C++, Shell script, Linux

Deep Learning Skills

- Libraries(PyTorch, TensorFlow, HuggingFace, VLLMs)
- Multi-GPU and multi-node training/inference

System & Infrastructure

- Linux Server Administration, Slurm-based Cluster Management

REFEREES

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Jeong-Uk Kim, Assistant Professor, Ph.D.
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